Weather and Climate Data Sharing in RA VI Region

WMO Unified Data Policy

Introduction – Group's action plan

- **Strategic Objective 1.2** – Broaden the provision of policy and decision-supporting climate information and services
  - **Task 2** - Facilitate implementation of WMO Unified Data Policy

Introduction - Unified Data Policy

- The **WMO Unified Data Policy** (Recommendation 3, EC-73) establishes the scope of WMO's policy on the international exchange of those Earth system data that are necessary for our efforts to monitor, understand and predict the weather and climate — including the hydrological cycle, the atmospheric environment and space weather.
- Provides a **single, comprehensive framework** for the free and unrestricted exchange of data across all WMO-relevant Earth system domains.
- **Consolidates and updates** previous separate data policies (such as Resolution 40 for weather, Resolution 25 for hydrology, and Resolution 60 for climate) into one overarching policy.

Introduction - Unified Data Policy

- **Adopted policy** on the international exchange of Earth system data:

  *As a fundamental principle of WMO and in consonance with the expanding requirements for its scientific and technical expertise, WMO commits itself to broadening and enhancing the **free and unrestricted** international exchange of Earth system data*

Introduction - Unified Data Policy

- **Two-Tiered Data Approach**:
- **Core Data:** Members **shall** exchange these data on a free and unrestricted basis. They are essential for protecting life and property.
- **Recommended Data:** Members **should** exchange these to further support Earth system monitoring and prediction at all levels.
- Encourages data exchange not just between National Meteorological and Hydrological Services (NMHSs), but also involves the private sector and academia.

- **Expanded Scope:** Covers seven key domains: **weather, climate**, hydrology, atmospheric composition, cryosphere, oceans, and space weather.

Introduction - Maturity matrix for climate data

- **The WMO Stewardship Maturity Matrix for Climate Data (SMM-CD)** is a tool designed for dataset owners to quantify and improve the management of climate data based on international best practices. It enables self-assessment across key areas like access, quality, and usability, supporting the modernization of data practices within National Meteorological and Hydrological Services (NMHSs).
- Purpose: To assess, rate, and improve the quality and management of climate datasets, particularly those in the WMO Catalogue for Climate Data.
- The matrix consists of four main categories for assessment:
- Data Access, Usability and Usage, Quality Management, Data Management

Introduction - Maturity matrix for climate data

Maturity scale levels for each aspect			
Level 1 (ad hoc) ☆	Level 2 (low) ☆☆	Level 3 (medium) ☆☆☆	Highly desirable ☆☆☆
Ad hoc	Intermediate	Advanced	Optimal
Not managed	Limit-managed	Managed	Level 3 plus additional featu (see text follow
Not implemented	Partially implemented	Fully implemented	

Figure 2. The maturity scale and structure for the SMM-CD_NRP

- **Structure of the SMM-CD_NRP**
- The SMM-CD has been adapted for national and regional data sets and a reduced number of maturity levels is used for the SMM-CD_NRP. The structure of the maturity scale of the SMM-CD_NRP

Data access matrix for weather and climate data

- **Structure of the DA matrix**
- Adopts **4 Maturity Levels** –
  - **Level 1 (ad hoc)**
  - **Level 2 (medium)**
  - **Level 3 (high)**
  - **Level 4 (advanced)**
- **4 Categories**
  - **Data access –** Discoverability, Data availability, Ease of access, Portal language
  - **Data portability**
  - **Documentation**
  - **Quality control**

Data access matrix for weather and climate data

- **Data access categories maturity levels**
- Discoverability –
  - **Data-set information not discoverable**; data not available publicly; person-to-person contact (e.g., email or formal request)
  - **Limited data-set information and metadata**; basic online services available for data access (e.g., ftp)
  - **Data set searchable online**; access provided with enhanced online data services
  - **Level 3 plus interoperable web services**; full capability of subsetting, aggregation and visualization, ability to download subset of data set

Data access matrix for weather and climate data

- **Data access categories maturity levels**
- Data availability –
  - **Only long-term archived data** (monthly/annual summaries); no real-time or recent updates
  - **Daily time-step data available** (e.g., temperature, precipitation), but no hourly or real-time feeds
  - **Hourly or sub-daily data available with short delay**; includes recent observations and forecasts
  - **Comprehensive access to both real-time feeds** (e.g., API, live dashboards) **and historical archives**; includes forecasts, radar, satellite, and station-level data

Data access matrix for weather and climate data

- **Data access categories maturity levels**
- Ease of access –
  - **Restricted / Technical Barrier** (Data access requires direct contact or technical API knowledge; no GUI or download options)
  - **Basic Access / Manual Download** (Data is available via downloadable files (e.g., Excel, CSV), but lacks search tools, previews, or filtering)
  - **Interactive Portal** (Web interface allows filtering, previewing, and downloading datasets; documentation is available)
  - **Advanced / User-Friendly Platform** (Full-featured portal with APIs, visualizations, dashboards, multilingual support, and tutorials)

Data access matrix for weather and climate data

- **Data access categories maturity levels**
- Language –
  - **National language/s only**
  - **National language/s plus partially English**
  - **National language/s plus English**
  - **National language/s plus multiple foreign languages**

Data access matrix for weather and climate data

- **Data portability maturity levels**
  - **Non-machine readable**
  - **Basic machine readable** (e.g., ASCII, with data and metadata header combined in single file)
  - **Standards-based machine readable**
  - **Machine independent, self-describing, interoperable format** (e.g., netCDF)

Data access matrix for weather and climate data

- **Documentation maturity levels**
  - **Product information not publicly available online**
  - **Limited online documentation** (e.g., user guide)
  - **Comprehensive documentation available online**
  - **Level 3 plus available in standard format template** (e.g., INSPIRE, NASA ATBD's)
  - Possible note – „only in national language“

Data access matrix for weather and climate data

- **Quality control**

  - **No QA or QC procedures available online**
  - **QA and QC procedures defined and available online**
  - **Level 2 plus data quality assessed and available online**, such as error or uncertainty estimates;
  - **Level 3 plus data-set citations in scientific publications in peer-reviewed journals**. Used/cited by other well-known climate data centers (NOAA/NCEI, DWD, etc.),

  - Possible note – „only in national language“

Data access matrix for weather and climate data

- **Compliance with WMO Unified Data Policy**
  - **Unclear or limited**
  - **Partial**
  - **Strong alignment**

Data access matrix for weather and climate data

- **Information from DA matrix are aggregated in a final dashboard**
  - Country name
  - Agency official
  - Agency‘s website
  - Open data link
  - Portal language
  - Discoverability
  - Data availability
  - Ease of access
  - Data portability
  - Documentation
  - Quality control
  - Compliance with WMO Unified Data Policy
  - Notes

Data access matrix for weather and climate data

region VI Europe						Data access						
Country	Agency official	Agency Website	Data link	Portal language	Discoverability	Data availability	Ease of access (simple)	Data portability	Documentation	Quality control	Compliance with WMO Unified Data Policy	Remarks
Albania	Institute of GeoSciences, Energy, Water and Environment	Institute of GeoSciences			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	Open Data Albania exists, but lacks meteorological datasets
Andorra	National Meteorological Service	Meteo ad. Servei Meteorològic Nacional.			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	The Servei Meteorològic Nacional, with the aim of Open Data, provides his weather forecaster in numerical format data with Json method
Armenia	Hydrometeorology and Monitoring Center" (HMC)	Հիդրոմետ			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	
Austria	GeoSphere Austria	Wetterprognose Österreich	GeoSphere Austria Data Hub	National language/s only	Level 2 plus comprehensive discovery metadata: data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Advanced / User-Friendly Platform	Standards-based machine readable	Comprehensive documentation available online	Level 3 plus data-set citations in scientific publications in peer-reviewed journals, liked/cited by other well-known climate data centres (NOAA, NCEI, DWD, etc.).	Strong alignment	Open Data Armenia exists, but lacks core meteorological datasets. [1]
Azerbaijan	National Hydrometeorological Department	Azərbaycan Respublikası Ekologiya və Təbii Sərvətlər Nazirliyi / Азербайджан Республикası Экология və Təbii Sərvətlər Nazirliyi			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	member of RODED initiative
Belarus	Belhydromet	Agrohydrometeorology	Belhydromet - Main page		Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	
Belgium	Institut royal météorologique	Météo en Belgique - RMI	Data	Belgian Climate Centre: https://opendata.meteo.be/home	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Advanced / User-Friendly Platform	Standards-based machine readable	Comprehensive documentation available online	QA and QC procedures defined and available online	Strong alignment	member of RODED initiative
Bulgaria	National Institute of Meteorology and Hydrology (NIMH)	meteo.bg			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Partial	NIMH data 3rd party site - https://meteo.bg/ National Open Data portal exists - https://data.egov.bg/
Croatia	Croatian Meteorological and Hydrological Service (CMHMZ)	DHMZ - Croatian Meteorological and Hydrological Service			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Partial	
Cyprus	Department of Meteorology of the Republic of Cyprus	Home Page | Department of Meteorology	Department of Meteorology | National Opendata Portal	National language/s plus partially English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Hourly or sub-daily data available with short delay; includes recent observations and forecasts	Restricted / Technical Barrier	Standards-based machine readable	Comprehensive documentation available online	No QA or QC procedures defined and available online	Partial	
Czechia	Czech Hydrometeorological Institute (CHMI)	Portál CHMÚ - Home	CHMI Open Data	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Hourly or sub-daily data available with short delay; includes recent observations and forecasts	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	
Denmark	Danish Meteorological Institute (DMI)	Forside | DMI	DMI Open Data | DMI Open Data	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online	Level 2 plus data quality assessed and available online, such as error or uncertainty estimates;	Strong alignment	member of RODED initiative
Estonia	Estonian Environment Agency (EESTEA)	Estonian Weather Service	Climate data | Environment Portal https://www.ilmateenistus.ee/kliima/andmed-ilmaandmed/	National language/s plus partially English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Hourly or sub-daily data available with short delay; includes recent observations and forecasts	Restricted / Technical Barrier	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	
Finland	Finnish Meteorological Institute	Homepage - Finnish Meteorological Institute	Open data - Finnish Meteorological Institute	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	member of RODED initiative, https://eumetnet.eu/observations/observations-data-sharing
France	Météo-France	Site Officiel de Météo-France	https://portail-api.meteofrance.fr/web/en/	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Machine-independent, self-describing, interoperable format [e.g., netCDF]	Comprehensive documentation available online	QA and QC procedures defined and available online	Strong alignment	Governmental open data portal is only in French
France			Observation data from the Météo-France ground-based observation network - ceda	National language/s plus English								
France			Accueil | meteo.data.gouv.fr	National language/s only								
France			METEO FRANCE Data Playground	National language/s plus English								
Georgia	National Hydrometeorological Service/ National Environmental Agency	National Environmental Agency | Nea			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	
Germany	Deutscher Wetterdienst (DWD)	Wetter und Klima - Deutscher Wetterdienst - Startseite	Wetter und Klima - Deutscher Wetterdienst - Climate information systems	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online	Level 2 plus data quality assessed and available online, such as error or uncertainty estimates;	Strong alignment	Observations Data Sharing - Eumetnet
Greece	Hellenic National Meteorological Service (HNMS)	HNMS - Home	Meteorological data time series (Daily) - Datasets - HELIX	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Daily time-step data available [e.g., temperature, precipitation], but no hourly or real-time feeds	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	
Hungary	Hungarian Meteorological Service (HungaroMet Nonprofit Zrt.)	IDŐJÁRÁS - met.hu	Climate of Hungary - met.hu	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Daily time-step data available [e.g., temperature, precipitation], but no hourly or real-time feeds	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	member of RODED initiative
Iceland	The Icelandic Meteorological Office	https://en.vedur.is/	https://en.vedur.is/climatology/data/	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Only long-term archived data (monthly/annual summaries); no real-time or recent updates	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	member of RODED initiative
Ireland	The Irish Meteorological Service (Met Éireann)	https://www.met.ie/	Public Service Data Catalogue.	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online	No QA or QC procedures defined and available online	Strong alignment	Met Eireann partially functional open data portal
Ireland			Met Eireann - Publishers - data.gov.ie									
Israel	Israel Meteorological Service	Home | Israel Meteorological Service	Meteorological Database	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online (ONLY IN NATIONAL LANGUAGE)	QA and QC procedures defined and available online (ONLY IN NATIONAL LANGUAGE)	Strong alignment	FileCloud > Examplar_Unified Data Policy Implementation_ISRAEL_Oct 2022.pdf
Italy	Italian Air Force Meteorological Service	Meteo Aeronautica Militare	meteohub	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Hourly or sub-daily data available with short delay; includes recent observations and forecasts	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online	No QA or QC procedures defined and available online	Strong alignment	
Italy			INSPIRE	National language/s plus English								
Kazakhstan	Kazhydromet	https://www.kazhydromet.kz/en/			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]			Non-machine readable	Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	
Lebanon	Météo Liban	METEO Liban - Official Weather Services - Lebanon	Weather statistics 1996-2016 - Open Data Lebanon	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Only long-term archived data (monthly/annual summaries); no real-time or recent updates	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	
Latvia	Latvian Environment, Geology and Meteorology Centre (LEGMC)	LVGMC	Weather in Latvia by selected cities and towns 1997 - 2024. PxWeb	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Only long-term archived data (monthly/annual summaries); no real-time or recent updates	Interactive Portal	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	member of RODED initiative
Lithuania	Lithuanian Hydrometeorological Service	Meteo.lt - Lietuvos hidrometeorologijos tarnyba	Lietuvos atvirų duomenų portalas | Resources	National language/s plus partially English	Level 2 data set searchable online; access provided with enhanced online data services	Hourly or sub-daily data available with short delay; includes recent observations and forecasts	Interactive Portal	Standards-based machine readable	Comprehensive documentation available online	No QA or QC procedures defined and available online	Strong alignment	
Luxembourg		meteolux	Meteorological data - Portail Open Data	National language/s plus English	Limited data-set information and discovery metadata; basic online services available for data access [e.g., ftp]	Daily time-step data available [e.g., temperature, precipitation], but no hourly or real-time feeds	Basic Access / Manual Download	Standards-based machine readable	Limited online documentation [e.g., user guide]	No QA or QC procedures defined and available online	Partial	
Malta												
Monaco												
Montenegro	Institute of Hydrometeorology and Seismology of Montenegro	https://www.meteo.co.me/index.php			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]				Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	
Netherlands (Kingdom of the)	Royal Netherlands Meteorological Institute (KNMI)	https://www.knmi.nl/home	https://dataplatform.knmi.nl/	National language/s plus English	Level 2 data set searchable online; access provided with enhanced online data services	Comprehensive access to both real-time feeds [e.g., API, live dashboards] and historical archives; includes forecasts, radar, satellite, and station-level data	Interactive Portal	Machine-independent, self-describing, interoperable format [e.g., netCDF]	Comprehensive documentation available online	QA and QC procedures defined and available online	Full alignment	member of RODED initiative, https://eumetnet.eu/observations/observations-data-sharing
North Macedonia	Hydrometeorological Service (HMS), North Macedonia	https://vhm.gov.mk/			Data-set information not discoverable; data not available publicly; person-to-person contact [e.g., email or formal request]				Product information not publicly available online	No QA or QC procedures defined and available online	Unclear or limited	main weather only in national language

Data access matrix for weather and climate data

- **Preliminary results** from 43 of 55 countries (78 %)
  - **23 countries** / **NWS** (**53 %**) provided dedicated portals/ links to open data portals
    - **19 portals (83 %)** are available in **English**
    - **13 portals (57 %)** met criteria for **level 3** for **discoverability** - data set searchable online; access provided with enhanced online data services
    - **16 portals (70 %)** met criteria for **level 3** for **data availability** - Hourly or sub-daily data available with short delay; includes recent observations and forecasts
    - **23 portals (100 %)** met criteria for **level 3** for **Data portability -** Standards-based machine readable
    - **12 portals (52 %)** met criteria for **level 3** for **Documentation -** Comprehensive documentation available online
    - **9 portals (40 %)** met criteria for **level 2** for **Quality Control -** QA and QC procedures defined and available online
  - **11 countries / NWS** (**48 %**) were evaluated as **„strongly aligned“** with WMO Unified Data Policy

Data access matrix for weather and climate data

- **Example of the Dashboard assessment -** Header
  - **RA VI Country name**
    - Denmark
  - **Agency official**
    - Danish Meteorological Institute (DMI)
  - **Agency's website**
    - https://www.dmi.dk/
  - **Open data link**
    - https://www.dmi.dk/friedata/dokumentation/data

Data access matrix for weather and climate data

- **Example of the Dashboard assessment – Data access matrix**
  - **Portal language**
    - National language/s plus English
  - **Discoverability**
    - Data set searchable online; access provided with enhanced online data services
  - **Data availability**
    - Comprehensive access to both real-time feeds (e.g., API, live dashboards) and historical archives; includes forecasts, radar, satellite, and station-level data
  - **Ease of access**
    - Interactive Portal

Data access matrix for weather and climate data

- **Example of the Dashboard assessment - matrix**
  - **Data portability**
    - Standards-based machine readable
  - **Documentation**
    - Comprehensive documentation available online
  - **Quality control**
    - Level 2 plus data quality assessed and available online, such as error or uncertainty estimates
  - **Compliance with WMO Unified Data Policy**
    - Strong alignment

Next Steps & Path Forward

- Validation and Expert Review
  - **Refine Assessment Criteria:** Seek feedback to validate the **Data Access (DA) Matrix** categories and maturity level thresholds.
  - **Expand the Dashboard:** Apply the maturity assessment to all remaining NMHSs within the RA VI region.
- Implementation Support
  - **Broaden Stakeholder Engagement**
  - **Address Identified Gaps**